

Retrieval-Based Adaptation of Time Series Foundation Models

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Abstract

This report describes my current work on retrieval-based adaptation for pretrained time series forecasting models. We briefly introduce notations, the literature context, and the implemented baselines as well as the proposed TS-IFA architecture.

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1 Time Series Forecasting

Let $\mathcal{U} = \{0, \dots, U-1\}$ be a set of U users or channels and $t \in \{0, \dots, T-1\}$ the time index for the T measured dates. For channel $u \in \mathcal{U}$, the univariate series is $z_0^{(u)}, \dots, z_{T-1}^{(u)}$. We consider the setting of a fixed lookback length $L \geq 1$ and forecast horizon $H \geq 1$. Given a valid *query date* s , the lookback and future horizon are

$$X_{s,u} = z_{(s-L,s]}^{(u)} = (z_{s-L+1}^{(u)}, \dots, z_s^{(u)}) \in \mathbb{R}^L, \quad (1)$$

$$Y_{s,u} = z_{(s,s+H]}^{(u)} = (z_{s+1}^{(u)}, \dots, z_{s+H}^{(u)}) \in \mathbb{R}^H. \quad (2)$$

Thus the lookback length L represents the number of observed values at a given time and the horizon H is the number of target values to predict.

A forecasting model is a parameterized map

$$F_\theta : \mathbb{R}^L \longrightarrow \mathbb{R}^H, \quad X_{s,u} \longmapsto \hat{Y}_{s,u}.$$

Here θ denotes the model parameters.

Certain families of models allow the use of optional covariates. Let $C_{s,u}$ denote a set of covariate windows of size $L+H$ (horizon included). When using covariates, the prediction becomes $\hat{Y}_{s,u}^{\text{cov}} = F_\theta(X_{s,u}; C_{s,u})$.

For any $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, define

$$\mu_x = \frac{1}{n} \sum_{k=1}^n x_k, \quad \sigma_x = \text{sd}(x) = \left(\frac{1}{n} \sum_{k=1}^n (x_k - \mu_x)^2 \right)^{1/2}, \quad \varepsilon = 10^{-8}.$$

For one sample, consisting of windows $(X, Y, \hat{Y})$, the principal losses are

$$\text{MSE}(\hat{Y}, Y) = \frac{1}{H} \sum_{h=1}^H (\hat{Y}_h - Y_h)^2, \quad \text{nMSE}(\hat{Y}, Y \mid X) = \frac{1}{H} \sum_{h=1}^H \left(\frac{\hat{Y}_h - Y_h}{\max(\sigma_X, \varepsilon)} \right)^2.$$

2 Foundation Models

A time series Foundation Model (FM) is pretrained over many tasks and datasets, and directly reused on unseen series without updating its parameters. Indeed, their extensive training allows them to perform In-Context Learning (ICL), which means the model has learned to detect patterns within the input, as if it were a dataset on which it was fine-tuned. A few examples of FMs are provided below.

Chronos-2. Chronos-2 is a 120M-parameter encoder-only foundation model whose group attention supports univariate, multivariate, and covariate-aware zero-shot forecasting (Ansari et al., 2025). In this project, we notably use context mode `future_included`: each covariate contributes its lookback as well as its observed horizon. The active code name is `chronos2`.

TabPFN-TS. TabPFN-TS casts forecasting as tabular regression with time, seasonal, frequency, and optional covariate features, then performs ICL with a pretrained TabPFN regressor (Hoo et al., 2025). The project wrapper uses the `tabpfm-v2.5-regressor-v2.5_default.ckpt` checkpoint, automatically adds periods 24 and 168 when L permits, and exposes the code name `tabpfnts`.

TS-ICL. TS-ICL is a 27M-parameter time-indexed encoder-regressor Transformer for forecasting and imputation on regular or irregular timestamps (Le Naour et al., 2026). It is part of the literature scope, but `ts_icl` is deliberately rejected by the current launcher because no wrapper is registered.

Chronos-Bolt. The `chronos-bolt` wrapper supplies the frozen 205M-parameter Chronos-Bolt-base backbone for the Cross-RAG comparison. It is univariate and ignores retrieved covariates; neighbour targets and forecasts remain available to the external adapter (Ansari et al., 2024; Amazon Science, 2026).

Table 1: Model names present in the Python registry or publication launchers. Trainable means parameters updated by this project, not total pretrained size.

Code name	Role and current status	Total parameters	Trainable here
chronos2	Primary screen/full backbone; frozen	120M	0
tabpfnts	Ultra backbone; frozen TabPFN-v2.5 regressor	≈ 11 M	0
chronos-bolt	Fixed Cross-RAG backbone; frozen	205M	0
persistence	Python-registry diagnostic	0	0
linear	Python-registry diagnostic, univariate	$H(L+1)$	$H(L+1)$
patchtst	Registry diagnostic; not launched for publication	466,843 / 663,091 / 1,762,603	same
ts_icl	Reserved shell name; not registered	27M in the paper	not run

The three PatchTST numbers in Table 1 use the code defaults at 168:24, 336:48, and 504:168, respectively. Publication experiments accept only the three frozen backbones listed first.

3 From RAG to retrieval-augmented forecasting

Retrieval-augmented generation separates a parametric model from an explicit, replaceable non-parametric memory. Classical RAG retrieves documents from a dense index and conditions a pretrained generator on them (Lewis et al., 2020). In k NN-MT, a frozen neural machine-translation model caches decoder representations k_i with next-token values v_i ; at decoding state q_t , it interpolates the model and neighbour distributions (Khandelwal et al., 2021). Let $p_\theta(\cdot | q_t)$ be the neural next-token distribution for candidate token y_t , let $\mathcal{N}_K(q_t)$ contain the $K \geq 1$ cached keys nearest to q_t under distance d , and choose interpolation weight $\lambda \in [0, 1]$ and temperature $\tau > 0$. Then

$$p(y_t | q_t) = (1 - \lambda)p_\theta(y_t | q_t) + \lambda \frac{\sum_{i \in \mathcal{N}_K(q_t)} \mathbf{1}[v_i = y_t] \exp(-d(q_t, k_i)/\tau)}{\sum_{i \in \mathcal{N}_K(q_t)} \exp(-d(q_t, k_i)/\tau)}.$$

This is the direct ancestor of our question: can a frozen pretrained model be adapted cheaply by retrieving domain examples and mixing their outcomes?

Time-series work has followed several paths: retrieved sequences can become a text prompt (TimeRAG), an appended context (RAF), an input to a model trained on the target dataset (RAFT), or values fused by pretrained attention modules (TS-RAG and Cross-RAG). Our method is narrower: the datastore is target-panel history, the backbone is frozen, and simple validated forecasts or gates are tested before a neural adapter.

Table 2: Bibliography map for pretrained retrieval and time-series forecasting. This is a method map, not a cross-paper accuracy ranking.

Method	Date / venue	Retrieval object	Adaptation mechanism
RAG (Lewis et al., 2020)	2020, NeurIPS	Dense-indexed passages	Fine-tuned seq2seq generator combines parametric and non-parametric memory.
k NN-MT (Khandelwal et al., 2021)	2021, ICLR	Decoder state / next-token pairs	No retraining; interpolates neural and neighbour token distributions.
RAF (Tire et al., 2024)	2024 preprint; v2 2026	Related time-series contexts and futures	Normalises, aligns, and prepends retrieved examples to frozen TSFMs.
TimeRAG (Yang et al., 2025)	2024 preprint; ICASSP 2025	DTW-similar historical sequences	Renders references and query as an LLM forecasting prompt.
TS-RAG (Ning et al., 2025a)	2025, NeurIPS	Encoder-space context-future pairs	Pretrains projectors and an Adaptive Retrieval Mixer; no target-task tuning.
RAFT (Han et al., 2025)	2025, ICML	Correlated training inputs and following patches	Trains a shallow multiresolution forecasting MLP per target dataset.
Cross-RAG (Lee et al., 2026a)	2026, KDD MILETS	Min-max lookback/future pairs, cosine search	Pretrained query-retrieval cross-attention with query-dependent and query-independent paths.
This project	2026	Raw or transformed historical lookback/future pairs	Closed-form baselines, CatBoost gates, and TS-IFA around a frozen backbone.

4 Neighbour retrieval

4.1 Datastore and exact search

Fix a named retrieval transform ρ and distance metric δ . Their maps are

$$\phi = \phi_\rho : \mathbb{R}^L \rightarrow \mathbb{R}^{d_\rho}, \quad d = d_\delta : \mathbb{R}^{d_\rho} \times \mathbb{R}^{d_\rho} \rightarrow \mathbb{R}_{\geq 0},$$

with every implemented ϕ_ρ and d_δ defined explicitly in Table 3. Here $d_\rho = L$ except for the encoder space; $g_\theta : \mathbb{R}^L \rightarrow \mathbb{R}^{d_{\text{enc}}}$ denotes the frozen backbone representation after the configured flattening or pooling.

For query (s, u) , let $\mathcal{R}_s = \{r_{s,0} < \dots < r_{s,n_s^{\text{store}}-1}\}$ be the ordered finite set of $n_s^{\text{store}} = |\mathcal{R}_s|$ store dates in $\{L-1, \dots, T-H-1\}$ that pass the fixed/online, period, stride, and cap rules in Section 7. The eligible store keys and the resulting labelled datastore are

$$\mathcal{I}_s = \mathcal{R}_s \times \mathcal{U}, \quad \mathcal{D}_s = \{(\phi(X_{r,v}), X_{r,v}, Y_{r,v}, r, v) : (r, v) \in \mathcal{I}_s\}.$$

For $K \in \{1, \dots, |\mathcal{I}_s|\}$, let $\text{TopK}^{\min}$ return the ordered K minimizers. Exact search returns the following store keys:

$$\mathcal{N}_K(s, u) = \text{TopK}_{(r,v) \in \mathcal{I}_s}^{\min} d(\phi(X_{s,u}), \phi(X_{r,v})).$$

The datastore is flattened in *user-major, date-minor* order. For $i \in \{0, \dots, Un_s^{\text{store}} - 1\}$, a flat index decodes as

$$v = \left\lfloor \frac{i}{n_s^{\text{store}}} \right\rfloor, \quad m = i \bmod n_s^{\text{store}}, \quad r = r_{s,m}.$$

Downstream query payloads use the opposite, date-major ordering $(s_0, u_0), \dots, (s_0, u_{U-1}), (s_1, u_0), \dots$ so that whole dates form contiguous blocks for chronological splitting; $s_0 < s_1 < \dots$ are the saved query dates and $u_k = k$ is the internal channel index.

In Table 3, $X \in \mathbb{R}^L$ is a generic lookback; in its distance rows, $a, b \in \mathbb{R}^{d_\rho}$ are two transformed lookbacks and $x_{\min} = \min_{1 \leq k \leq L} X_k$, $x_{\max} = \max_{1 \leq k \leq L} X_k$, $a^\circ = a - \mu_a \mathbf{1}_{d_\rho}$, $b^\circ = b - \mu_b \mathbf{1}_{d_\rho}$ are their centred forms.

Table 3: Implemented retrieval spaces and distances.

Space	$\phi(X)$	Status
Raw	X	Primary screen
Instance	$(X - \mu_X \mathbf{1}_L) / (\sigma_X + \varepsilon)$	Primary screen
Min-max	$(X - x_{\min} \mathbf{1}_L) / \max(x_{\max} - x_{\min}, \varepsilon)$	Cross-RAG profile
Fourier	$ \text{FFT}((X - \mu_X \mathbf{1}_L) / (\sigma_X + \varepsilon)) $	Optional ablation
Encoder	flattened or pooled frozen representation $g_\theta(X)$	Implemented override
Euclidean	$\ a - b\ _2$	Primary metric
Cosine	$1 - a^\top b / [\max(\ a\ _2, \varepsilon) \max(\ b\ _2, \varepsilon)]$	Cross-RAG metric
Pearson	$1 - (a^\circ)^\top b^\circ / [\max(\ a^\circ\ _2, \varepsilon) \max(\ b^\circ\ _2, \varepsilon)]$	Implemented override

4.2 Saved information and query-scale transfer

For neighbour rank $j \in \{1, \dots, K\}$ with $(r_j, v_j) \in \mathcal{N}_K(s, u)$, abbreviate $X_j = X_{r_j, v_j}$ and $Y_j = Y_{r_j, v_j}$, and let d_j be its retrieval distance and $N_j = F_\theta(X_j)$ its frozen forecast. Also abbreviate the query as $X = X_{s,u}$ and $Y = Y_{s,u}$. Extraction saves the quantities in Table 4; it stores $E_j = Y_j - N_j$ rather than duplicating N_j , which is reconstructed downstream.

Table 4: Current extraction payload. Optional fields are generated only when their switches are enabled.

Object	Saved content
Query	$X, Y, V = F_\theta(X), C = F_\theta(X; \{X_j, Y_j\}_{j=1}^K)$
Neighbours	$X_j, Y_j, E_j = Y_j - N_j, d_j, r_j, v_j$ for $j = 1, \dots, K$
Indices	query date/user, neighbour date/user, eligible store-date/window counts
Summaries	query and neighbour level/scale; forecast-context and residual losses
Optional	context-conditioned neighbour residuals

Retrieved values are expressed on the query scale before fitting. Write $q = (s, u)$ for the current query and let $(\mu_q, \sigma_q) = (\mu_X, \sigma_X)$ and $(\mu_j, \sigma_j) = (\mu_{X_j}, \sigma_{X_j})$. For a neighbour-scale horizon vector $Z \in \mathbb{R}^H$ or residual vector $E \in \mathbb{R}^H$, define

$$\mathcal{T}_{j \rightarrow q}(Z) = \frac{Z - \mu_j \mathbf{1}_H}{\max(\sigma_j, \varepsilon)} \sigma_q + \mu_q \mathbf{1}_H, \quad (3)$$

$$\mathcal{T}_{j \rightarrow q}^{\text{res}}(E) = \frac{E}{\max(\sigma_j, \varepsilon)} \sigma_q. \quad (4)$$

All later symbols Y_j, N_j, E_j denote these query-scale values.

Distances are converted to weights independently for every query. Let $\mathbf{d} = (d_1, \dots, d_K)$ be the ordered retrieved-distance vector and let $j, \ell \in \{1, \dots, K\}$:

$$\tilde{d}_j = \frac{d_j - \min_{\ell} d_{\ell}}{\max(\sigma_{\mathbf{d}}, \varepsilon)}, \quad w_j = \frac{e^{-\tilde{d}_j}}{\sum_{\ell=1}^K e^{-\tilde{d}_{\ell}}}, \quad \bar{Y}_h = \sum_{j=1}^K w_j Y_{j,h}.$$

Thus $w_j \geq 0$, $\sum_{j=1}^K w_j = 1$, and $\bar{Y} \in \mathbb{R}^H$; for $K = 1$, $w_1 = 1$.

5 Retrieval baselines

For extracted window i with query (s_i, u_i) , define the target $G_i = Y_{s_i, u_i} \in \mathbb{R}^H$, vanilla forecast $V_i \in \mathbb{R}^H$, conditioned forecast $C_i \in \mathbb{R}^H$, query-scale neighbour future and forecast $(Y_{i,j}, N_{i,j}) \in \mathbb{R}^H \times \mathbb{R}^H$ for $j \in \{1, \dots, K\}$, and

$$V_i = F_{\theta}(X_{s_i, u_i}), \quad C_i = F_{\theta}(X_{s_i, u_i}; \{X_{i,j}, Y_{i,j}\}_{j=1}^K), \quad \bar{Y}_i = \sum_{j=1}^K w_{i,j} Y_{i,j}.$$

Each design is $(V_i, Z_{i,1}, \dots, Z_{i,p})$.

Table 5: Baseline designs; $Z_i = (Z_{i,1}, \dots, Z_{i,p})$.

Stem	Z_i	p
cov	(C_i)	1
avgy	$(\bar{Y}_i)$	1
y	$(Y_{i,1}, \dots, Y_{i,K})$	K
cov_y	$(C_i, Y_{i,1}, \dots, Y_{i,K})$	$1 + K$
cov_avgy	$(C_i, \bar{Y}_i)$	2
residual	$(Y_{i,1}, \dots, Y_{i,K}, N_{i,1}, \dots, N_{i,K})$	$2K$
full	$(C_i, Y_{i,1}, \dots, Y_{i,K}, N_{i,1}, \dots, N_{i,K})$	$1 + 2K$

Let $\tau \in \{S, H\}$ denote shared or horizon-wise coefficients. For any coefficient vector c , set $c_h^S = c$ for every h , whereas $c_1^H, \dots, c_H^H$ are fitted independently. Define the simplex $\mathcal{S}_p = \{\lambda \in \mathbb{R}^{p+1} : \lambda_m \geq 0, \sum_{m=0}^p \lambda_m = 1\}$. The three families are

$$\hat{G}_{i,h}^{R,\tau} = V_{i,h} + a_h^{\tau} V_{i,h} + \sum_{m=1}^p b_{h,m}^{\tau} Z_{i,h,m}, \quad (a_h^{\tau}, b_{h,1}^{\tau}, \dots, b_{h,p}^{\tau}) \in \mathbb{R}^{p+1}, \quad (5)$$

$$\hat{G}_{i,h}^{C,\tau} = \lambda_{h,0}^{\tau} V_{i,h} + \sum_{m=1}^p \lambda_{h,m}^{\tau} Z_{i,h,m}, \quad \lambda_h^{\tau} \in \mathcal{S}_p, \quad (6)$$

$$\hat{G}_{i,h}^{D,\tau} = V_{i,h} + \sum_{m=1}^p b_{h,m}^{\tau} (Z_{i,h,m} - V_{i,h}), \quad b_h^{\tau} \in \mathbb{R}^p. \quad (7)$$

Vanilla is recovered by $(a, b) = 0$, $\lambda = (1, 0, \dots, 0)$, or $b = 0$. Ridge contains every convex forecast through $a = \lambda_0 - 1$ and $b_m = \lambda_m$. For cov, the formulas are $V + aV + bC$, $\lambda V + (1 - \lambda)C$, and $V + b(C - V)$.

Let $\mathcal{Q}_A$ be the windows in split $A \in \{T_1, T_2, T_{12}\}$, where $T_{12} = T_1 \cup T_2$. For $\tau = S$, set $\mathcal{O}_A^S = \mathcal{Q}_A \times \{1, \dots, H\}$; for the h th horizon fit, set $\mathcal{O}_A^{H,h} = \mathcal{Q}_A \times \{h\}$. Ridge and delta-ridge use correction target $t_{i,h} = G_{i,h} - V_{i,h}$ and design

$$d_{i,h}^R = (V_{i,h}, Z_{i,h,1}, \dots, Z_{i,h,p}) \in \mathbb{R}^{p+1}, \quad d_{i,h}^D = (Z_{i,h,1} - V_{i,h}, \dots, Z_{i,h,p} - V_{i,h}) \in \mathbb{R}^p.$$

For either $d \in \mathbb{R}^q$ and an observation set $\mathcal{O}$, define

$$r_k = \max \left(\sqrt{\frac{1}{|\mathcal{O}|} \sum_{(i,h) \in \mathcal{O}} d_{i,h,k}^2}, 10^{-12} \right), \quad \tilde{d}_{i,h,k} = d_{i,h,k}/r_k,$$

and, for $\alpha \geq 0$,

$$\hat{\gamma}_{\alpha, \mathcal{O}} = \arg \min_{\gamma \in \mathbb{R}^q} \frac{1}{|\mathcal{O}|} \sum_{(i,h) \in \mathcal{O}} (t_{i,h} - \tilde{d}_{i,h}^\top \gamma)^2 + \alpha \|\gamma\|_2^2, \quad \hat{\beta}_{\alpha, \mathcal{O}, k} = \hat{\gamma}_{\alpha, \mathcal{O}, k} / r_k.$$

The grid is

$$\mathcal{A} = \{0, 10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}.$$

For each $\alpha \in \mathcal{A}$, coefficients fitted on T_1 are scored by nMSE on T_2 ; the minimizer (smallest α on ties) is refitted on T_{12} . Convex coefficients are fitted directly on T_{12} :

$$\hat{\lambda}_{\mathcal{O}} = \arg \min_{\lambda \in \mathcal{S}_p} \frac{1}{|\mathcal{O}|} \sum_{(i,h) \in \mathcal{O}} \left(G_{i,h} - \lambda_0 V_{i,h} - \sum_{m=1}^p \lambda_m Z_{i,h,m} \right)^2.$$

Use $\mathcal{O} = \mathcal{O}_A^S$ for a shared fit and $\mathcal{O} = \mathcal{O}_A^{H,h}$ independently for each h ; no intercept or centring is added. Chunked sufficient statistics give the exact objectives.

Table 6: Baseline names; substitute a stem from Table 5.

Artifact	Definition	Fitted scalars
vanilla	$V_{i,h}$	0
cov_forecast	$C_{i,h}$	0
avgy	$\bar{Y}_{i,h}$	0
y_mean	$K^{-1} \sum_j Y_{i,j,h}$	0
<stem>_ridge_*	Equation (5), * = shared/horizon	$(p+1) / H(p+1)$
<stem>_convex_*	Equation (6), * = shared/horizon	$(p+1) / H(p+1)$
<stem>_delta_ridge_*	Equation (7), * = shared/horizon	p / Hp

6 Gates and features

A gate compares V_i with $A_i = C_i$ (cov) or $A_i = \bar{Y}_i$ (avgy). Define the pointwise and shared candidate advantages

$$\Delta_{i,h} = (G_{i,h} - V_{i,h})^2 - (G_{i,h} - A_{i,h})^2, \quad \bar{\Delta}_i = H^{-1} \sum_{h=1}^H \Delta_{i,h}.$$

For $q = (q_1, \dots, q_K) \in \mathbb{R}^K$, define neighbour-weighted moments

$$\mu_{w_i}(q) = \sum_{j=1}^K w_{i,j} q_j, \quad \sigma_{w_i}(q) = \left[\sum_{j=1}^K w_{i,j} (q_j - \mu_{w_i}(q))^2 \right]^{1/2}.$$

Let μ_h and σ_h denote population moments over $h \in \{1, \dots, H\}$. For neighbour j , set

$$\begin{aligned} r_{i,j}^V &= \mu_h(Y_{i,j,h} - V_{i,h}), & s_{i,j}^V &= \sigma_h(Y_{i,j,h} - V_{i,h}), \\ r_{i,j}^N &= \mu_h(Y_{i,j,h} - N_{i,j,h}), & s_{i,j}^N &= \sigma_h(Y_{i,j,h} - N_{i,j,h}). \end{aligned}$$

Let $(r_{i,j}, v_{i,j})$ be neighbour j 's date and channel, and define

$$\begin{aligned} \ell_i &= (\mu_{X_{i,1}}, \dots, \mu_{X_{i,K}}), & \rho_i &= (\sigma_{X_{i,1}}, \dots, \sigma_{X_{i,K}}), & \mathbf{a}_i &= (s_i - r_{i,1}, \dots, s_i - r_{i,K}), \\ \mathbf{w}_i &= (w_{i,1}, \dots, w_{i,K}), & \mathbf{d}_i &= (d_{i,1}, \dots, d_{i,K}). \end{aligned}$$

The exact static vector $f_i^0 \in \mathbb{R}^{18}$ is

$$\begin{aligned} f_i^0 &= (\mu_{w_i}(\mathbf{r}_i^V), \sigma_{w_i}(\mathbf{r}_i^V), \mu_{w_i}(\mathbf{s}_i^V), \mu_{w_i}(\mathbf{r}_i^N), \sigma_{w_i}(\mathbf{r}_i^N), \mu_{w_i}(\mathbf{s}_i^N), \\ &\quad \mu_{X_{s_i, u_i}}, \sigma_{X_{s_i, u_i}}, \mu_{\ell_i}, \sigma_{\ell_i}, \mu_{\rho_i}, \sigma_{\rho_i}, K^{-1} \sum_{j=1}^K \mathbf{1}[v_{i,j} = u_i], \\ &\quad \mu_{\mathbf{a}_i}, \sigma_{\mathbf{a}_i}, \sigma_{\mathbf{w}_i}, \max_j w_{i,j}, \mu_{\mathbf{d}_i}), \end{aligned}$$

where $\mathbf{r}_i^V = (r_{i,j}^V)_{j=1}^K$ and analogously for $\mathbf{s}_i^V, \mathbf{r}_i^N, \mathbf{s}_i^N$. The shared and horizon feature vectors are

$$f_i^S = (\mu_h(A_{i,h} - V_{i,h}), \sigma_h(A_{i,h} - V_{i,h}), f_i^0) \in \mathbb{R}^{20}, \quad (8)$$

$$f_{i,h}^H = (A_{i,h} - V_{i,h}, \mu_{w_i}((Y_{i,j,h} - V_{i,h})_j), \sigma_{w_i}((Y_{i,j,h} - V_{i,h})_j), \mu_{w_i}((Y_{i,j,h} - N_{i,j,h})_j), \sigma_{w_i}((Y_{i,j,h} - N_{i,j,h})_j), f_i^0) \in \mathbb{R}^{23}. \quad (9)$$

For shape $\tau \in \{S, H\}$, write $z_i^S = \bar{\Delta}_i$, $z_{i,h}^H = \Delta_{i,h}$, f_i^S as in Equation (8), and $f_{i,h}^H$ as in Equation (9). A shared model is fitted once; a horizon model is fitted independently for each h . CatBoost scores are

$$g^{\text{reg},\tau} = \text{CBReg}(f^\tau), \quad g^{\text{cls},\tau} = \Pr_{\text{CBCls}}(z^\tau > 0 \mid f^\tau) - \frac{1}{2}.$$

The tree count is selected on T_2 after fitting on T_1 , then the model is refitted on T_{12} with that count. Define

$$q^{\text{hard},\tau} = \mathbf{1}[g^\tau > 0], \quad \hat{G}_{i,h}^{\text{hard},\tau} = (1 - q_{i,h}^{\text{hard},\tau})V_{i,h} + q_{i,h}^{\text{hard},\tau}A_{i,h},$$

with $q_{i,h}^{S,\tau} = q_i^{S,\tau}$. For soft output, let $\text{sigmoid}(x) = (1 + e^{-x})^{-1}$ and

$$s^S = \text{sd}((z_i^S)_{i \in \mathcal{Q}_{T_{12}}}), \quad s_h^H = \text{sd}((z_{i,h}^H)_{i \in \mathcal{Q}_{T_{12}}}).$$

Writing s^τ for the matching shared or h th-horizon scale,

$$p^{\text{cls},\tau} = g^{\text{cls},\tau} + \frac{1}{2}, \quad p^{\text{reg},\tau} = \begin{cases} \text{sigmoid}(g^{\text{reg},\tau}/s^\tau), & s^\tau > 10^{-12}, \\ 0, & s^\tau \leq 10^{-12}, g^{\text{reg},\tau} < 0, \\ \frac{1}{2}, & s^\tau \leq 10^{-12}, g^{\text{reg},\tau} = 0, \\ 1, & s^\tau \leq 10^{-12}, g^{\text{reg},\tau} > 0, \end{cases}$$

$$\hat{G}_{i,h}^{\text{soft},\tau} = (1 - p_{i,h}^\tau)V_{i,h} + p_{i,h}^\tau A_{i,h}.$$

For shared output, $p_{i,h}^{S,\tau} = p_i^{S,\tau}$ for every h . The no-feature Bayes scores and scored-split oracle scores are

$$g^{B,S} = |\mathcal{Q}_{T_{12}}|^{-1} \sum_{i \in \mathcal{Q}_{T_{12}}} \bar{\Delta}_i, \quad g_h^{B,H} = |\mathcal{Q}_{T_{12}}|^{-1} \sum_{i \in \mathcal{Q}_{T_{12}}} \Delta_{i,h}, \quad g_i^{O,S} = \bar{\Delta}_i, \quad g_{i,h}^{O,H} = \Delta_{i,h},$$

and use the hard rule above.

Table 7: Gate names for $A \in \{\text{cov}, \text{avgy}\}$; $\tau = S/H$.

Name pattern	Definition	Fitted scalars / proxy
bayes_<A>_shared/horizon	$g^{B,\tau}$, hard	1 / H
catboost_<A>_regressor_shared/horizon	$g^{\text{reg},\tau}$, hard	≤ 6000 / $\leq 6000H$
catboost_<A>_classifier_shared/horizon	$g^{\text{cls},\tau}$, hard	≤ 6000 / $\leq 6000H$
catboost_<A>_*_shared/horizon_soft	same fit, soft output	0 additional
oracle_<A>_shared/horizon	$g^{O,\tau}$, hard	0

CatBoost uses at most 300 depth-4 trees, learning rate 0.03, 50-round early stopping, and balanced classifier classes (Prokhorenkova et al., 2018). The proxy is $300(16 \text{ leaf} + 4 \text{ split}) = 6000$. Screen uses only hard shared regressors; classifier, soft-regressor, and horizon-regressor forms are one-axis ablations.

7 Practical protocol

7.1 Chronological regions and window eligibility

For the T dates introduced in Section 1, extraction computes exclusive boundaries, where round means nearest-integer rounding and $[a, b) = \{a, a+1, \dots, b-1\}$ for integer dates:

$$b_0 = \text{round}(0.30T), \quad b_{12} = \text{round}(0.80T), \quad b_3 = T.$$

The extraction regions are the fixed datastore region $T_0 = [0, b_0)$, pooled adaptation region $T_{12} = [b_0, b_{12})$, and held-out evaluation region $T_3 = [b_{12}, T)$. For a target period $[b, e)$, eligible query dates are

$$s \in [\max(L-1, b-1), \min(T-H-1, e-H-1)]$$

at the configured query stride. Hence the first target may start at b with query index $b - 1$; its lookback may cross the boundary, but its complete target must be inside $[b, e)$. Changing L therefore preserves T_3 target dates whenever both runs remain eligible.

Each downstream learner re-splits T_{12} by unique query date. If $n_{12} \geq 2$ dates are present and the validation fraction is $\rho = 0.2$,

$$n_2 = \min(n_{12} - 1, \max(1, \text{round}(\rho n_{12}))), \quad n_1 = n_{12} - n_2.$$

The first n_1 and last n_2 whole dates form T_1 and T_2 , respectively, so $T_{12} = T_1 \cup T_2$ and the two sets are disjoint; users on the same date are never split.

7.2 Dynamically aligned datastores

Let P be the retrieval period and D_{stride} the datastore stride, with $P \mid D_{\text{stride}}$. Before optional explicit history bounds, define the admissible endpoint pair

$$(a_s, b_s) = \begin{cases} (L - 1, b_0 - H - 1), & \text{fixed mode,} \\ (L - 1, s - H), & \text{online mode.} \end{cases}$$

The first phase-aligned candidate is $r_s^{\text{align}} = a_s + ((s - a_s) \bmod P)$, and the uncapped date grid is

$$\mathcal{G}_s = \{r_s^{\text{align}} + \ell D_{\text{stride}} : \ell \in \mathbb{N}_0, r_s^{\text{align}} + \ell D_{\text{stride}} \leq b_s\}.$$

Let $\text{Tail}_M(\mathcal{G})$ denote the M most recent ordered elements of $\mathcal{G}$, with $\text{Tail}_{+\infty}(\mathcal{G}) = \mathcal{G}$. Let M_s be the minimum of the active date cap, the window-derived cap $\lfloor W_{\text{max}}/U \rfloor$, and, in default online mode, the number of dates in the fixed-mode grid for the same s ; an absent cap is $+\infty$, and `full_online_history` omits the last term. The code returns

$$\mathcal{R}_s = \text{Tail}_{M_s}(\mathcal{G}_s).$$

Thus every $r \in \mathcal{R}_s$ satisfies

$$(s - r) \bmod P = 0.$$

In fixed mode, both $(r - L, r]$ and $(r, r + H]$ lie in T_0 . In online mode,

$$r \geq L - 1, \quad r + H \leq s,$$

so the neighbour future is already observed. It may overlap the query lookback, but never the query target.

Without an explicit override, the code first counts the phase-aligned dates that a fixed T_0 store would provide for the current query. The online store then keeps that many *most recent* eligible dates, subject to the 30,000-window cap. Thus the store slides forward without growing merely because the query is later. `full_online_history` removes this implicit date-count cap; explicit date and window caps still apply. Defaults are $P = 96$ samples for ETTm1, ETTm2, ETT_T_15T, and ETT_L_15T, and $P = 24$ otherwise. An implicit datastore stride is rounded upward to a multiple of P .

7.3 Fitting and leakage controls

Table 8: What each chronological region may influence.

Family	T_1	T_2	T_3
Convex mixtures	diagnostic fit	final fit on pooled T_{12}	score once
Ridge baselines	fit all α	select α ; refit pooled	score once
Fixed-candidate CatBoost	fit up to 300 trees	select tree count; fresh pooled refit	score once
TS-IFA joint	optimize branches and one rooter	select and restore checkpoint	score once
TS-IFA meta	chronological support/query outer updates	fit active frozen-branch rooter	score once
Oracle diagnostics	—	—	target-aware appendix bound only

Ridge sufficient statistics are accumulated exactly in float64 chunks. Optional sample caps affect fitting only; final T_3 scoring always uses every saved sample. A future gate over a trainable candidate must train that candidate on T_1 and the gate on T_2 without pooled refitting, unless out-of-fold candidate predictions are introduced.

Dataset loaders discover `config.json` beside the selected CSV. Portable fields live at the JSON top level and adaptation-only overrides under `adaptation`. Project-scoped values override shared values, explicit run arguments override both, and `drop_users` is additive. The resolved path, content hash, and applied keys are recorded in the extraction manifest.

8 TS-IFA

TS-IFA is a lightweight neural adapter around the frozen backbone $F := F_\theta$. We again suppress (s, u) , so $X = X_{s,u}$ and $Y = Y_{s,u}$; all following values are on the query scale:

$$p = F(X), \quad p^c = F(X; \{X_j, Y_j\}_{j=1}^K), \quad n_j = F(X_j), \quad e_j = Y_j - n_j.$$

8.1 Candidate branches

Define the learned maps

$$\begin{aligned} A_r &: \mathbb{R}^{L+H} \times (\mathbb{R}^{L+H})^K \times (\mathbb{R}^H)^K \rightarrow \mathbb{R}^H, & M_r &: \mathbb{R}^{2H} \rightarrow \mathbb{R}^H, \\ A_m &: \mathbb{R}^L \times (\mathbb{R}^L)^K \times (\mathbb{R}^H)^K \rightarrow \mathbb{R}^H, & M_m &: \mathbb{R}^{2H} \rightarrow \mathbb{R}^H. \end{aligned}$$

A_r, A_m are multi-head cross-attention maps and M_r, M_m are correction MLPs. Residual attention compares query and neighbour *states*:

$$q_r = [X, p], \quad k_j^r = [X_j, n_j], \quad v_j^r = e_j, \quad (10)$$

$$z_r = A_r(q_r, \{k_j^r\}, \{v_j^r\}), \quad c_r = p + M_r([p, z_r]). \quad (11)$$

Memory attention reads realised neighbour horizons directly:

$$z_m = A_m(X, \{X_j\}, \{Y_j\}), \quad c_m = p + M_m([p, z_m]).$$

The default candidate bank is $\mathcal{C} = (c_0, c_1, c_2, c_3) = (p, p^c, c_r, c_m)$. The rooters exclude c_0 and operate on the three non-vanilla deltas $d_j = c_j - p$, $j \in \{1, 2, 3\}$. Fixed and learned transformed covariates are a separate experiment rather than TS-IFA candidates.

8.2 Neural and ridge rooters

The neural rooter receives no handcrafted distances or dispersion features. Its learned attention map is $A_g : \mathbb{R}^L \times (\mathbb{R}^L)^K \times (\mathbb{R}^L)^K \rightarrow \mathbb{R}^{d_g}$, where d_g is the configured representation width:

$$z_g = A_g(X, \{X_j\}, \{X_j\}) \in \mathbb{R}^{d_g}.$$

For the learned type embedding $\tau_j \in \mathbb{R}^{d_g}$ of delta d_j , with $j \in \{1, 2, 3\}$, let LN denote dimension-preserving layer normalization and let $S_\omega : \mathbb{R}^{2d_g+2H} \rightarrow \mathbb{R}^H$ be the shared MLP scorer. It returns the unconstrained horizon vector

$$a_j = S_\omega([\text{LN}(z_g), \text{LN}(p), \text{LN}(d_j), \text{LN}(\tau_j)]) \in \mathbb{R}^H, \quad (12)$$

$$\hat{Y}_{\text{neural}} = p + \sum_{j=1}^3 a_j \odot d_j, \quad a_j \in \mathbb{R}^H. \quad (13)$$

Here ω denotes all neural-rooter parameters: A_g , its layer normalizations, the type embeddings, and the shared scorer. There is no sigmoid, softmax, or convexity constraint. The vanilla fallback is $a_j = 0$ for every candidate.

The ridge rooter is horizon-wise on the same candidate deltas. In the joint variant its free coefficient matrix is $\nu \in \mathbb{R}^{3 \times H}$ and is updated by gradient descent with the branches. Only the meta variant uses the following closed-form solve. For any non-empty fit set $\mathcal{A}$, let $d_{i,j,h}$ be the horizon- h delta of candidate j on window i . Let $n_{\mathcal{A}} = |\mathcal{A}| \geq 1$. The differentiable meta-solve uses $\epsilon_{\text{meta}} = 10^{-8}$ and the final fit uses $\epsilon_{\text{fit}} = 10^{-12}$:

$$s_{j,h}^{(d,\mathcal{A})} = \begin{cases} \sqrt{n_{\mathcal{A}}^{-1} \sum_{i \in \mathcal{A}} d_{i,j,h}^2 + \epsilon_{\text{meta}}^2}, & \text{differentiable meta-solve,} \\ \max\left(\sqrt{n_{\mathcal{A}}^{-1} \sum_{i \in \mathcal{A}} d_{i,j,h}^2}, \epsilon_{\text{fit}}\right), & \text{final fit,} \end{cases} \quad \tilde{d}_{i,j,h} = d_{i,j,h} / s_{j,h}^{(d,\mathcal{A})}.$$

For ridge penalty $\alpha > 0$, the standardized coefficient matrix is

$$\Gamma_{\mathcal{A}}^*(\theta_b) = \arg \min_{\Gamma \in \mathbb{R}^{3 \times H}} \sum_{i \in \mathcal{A}} \sum_{h=1}^H \left(Y_{i,h} - p_{i,h} - \sum_{j=1}^3 \Gamma_{j,h} \tilde{d}_{i,j,h} \right)^2 + \alpha \|\Gamma\|_F^2, \quad \nu_{\mathcal{A},j,h}^*(\theta_b) = \Gamma_{\mathcal{A},j,h}^*(\theta_b) / s_{j,h}^{(d,\mathcal{A})},$$

where $\|\cdot\|_F$ is the Frobenius norm. The ridge prediction is $p_{i,h} + \sum_{j=1}^3 \nu_{j,h} d_{i,j,h}$. Both ridge variants have exactly $3H$ coefficients; $\alpha = 10^{-2}$ applies to meta-ridge solves only.

8.3 Training and initialization

Let θ_b collect the parameters of c_r and c_m , and let $\mathcal{B} = \{c_r, c_m\}$. For minibatch index set $\mathcal{A}$, active-rooter parameters $\psi \in \{\nu, \omega\}$, and coefficients $a_{i,j,h}$, define

$$\mathcal{L}_B^A(\theta_b) = \sum_{c \in \mathcal{B}} \text{nMSE}_{\mathcal{A}}(c, Y) + \lambda_V \sum_{c \in \mathcal{B}} \text{nMSE}_{\mathcal{A}}(c, p), \quad (14)$$

$$\mathcal{L}_R^A(\theta_b, \psi) = \text{nMSE}_{\mathcal{A}}(\hat{Y}, Y) + \lambda_V \text{nMSE}_{\mathcal{A}}(\hat{Y}, p) + \frac{\lambda_2}{3|\mathcal{A}|H} \sum_{i,j,h} a_{i,j,h}^2 + \frac{\lambda_S}{3|\mathcal{A}|(H-1)} \sum_{i,j,h < H} (a_{i,j,h+1} - a_{i,j,h})^2. \quad (15)$$

The last term is zero for $H = 1$. Active values are $\lambda_V = \lambda_2 = \lambda_S = 10^{-2}$ and $\lambda_B = 0.1$.

The two joint variants use all of T_1 :

$$(\theta_b^*, \nu^*) = \arg \min_{\theta_b, \nu} \mathcal{L}_R^{T_1}(\theta_b, \nu) + \lambda_B \mathcal{L}_B^{T_1}(\theta_b), \quad \text{joint ridge}, \quad (16)$$

$$(\theta_b^*, \omega^*) = \arg \min_{\theta_b, \omega} \mathcal{L}_R^{T_1}(\theta_b, \omega) + \lambda_B \mathcal{L}_B^{T_1}(\theta_b), \quad \text{joint neural}. \quad (17)$$

AdamW updates both parameter blocks. Joint ridge never uses a closed-form solve. The minimum-nMSE complete checkpoint on T_2 is restored, then scored once on T_3 ; no joint rooter is refitted on T_2 .

For meta learning, split T_1 by whole query dates into earlier support $\mathcal{S}_1$ and later query $\mathcal{Q}_1$. Meta ridge computes

$$\nu_e^*(\theta_b) = \nu_{\mathcal{S}_1^{(e)}}^*(\theta_b), \quad (18)$$

$$\theta_b^* = \arg \min_{\theta_b} \mathcal{L}_R^{\mathcal{Q}_1^{(e)}}(\theta_b, \nu_e^*(\theta_b)) + \lambda_B \mathcal{L}_B^{\mathcal{Q}_1^{(e)}}(\theta_b), \quad (19)$$

differentiating through each exact standardized 3×3 solve. It freezes θ_b^* and fits $\nu_{T_2}^*(\theta_b^*)$ in closed form. Meta neural starts from learned $\omega^{(0)} = \omega$ and takes

$$\omega^{(k+1)} = \omega^{(k)} - \eta_{\text{in}} \nabla_{\omega^{(k)}} \mathcal{L}_R^{\mathcal{S}_1^{(e)}}(\theta_b, \omega^{(k)}).$$

Its query loss updates θ_b and the initialization ω ; first-order MAML is the default. It then freezes θ_b and fits only ω on all T_2 . Both meta variants score T_3 once. The default query fraction is 0.2, $K_{\text{in}} = 1$, and $\eta_{\text{in}} = 10^{-3}$.

Output-zero initialization of M_r , M_m , the neural scorer, and the free joint-ridge matrix gives $c_r = c_m = p$, $a_j = 0$, and $\hat{Y} = p$ at step zero. All AdamW stages use learning rate 10^{-5} , weight decay 10^{-4} , batch size 256, and 10,000 one-random-batch epochs outside smoke mode. The exact variant, split, optimizer, regularizers, widths, caps, and seed enter the completion signature.

8.4 Parameter counts

The active configuration uses four attention heads, per-head dimension 32, 128-unit hidden layers, and three rooter deltas. K affects compute and memory, not parameter count.

Table 9: Exact current TS-IFA parameter counts, reproduced by the implementation and synchronized `config.json` artifacts.

L	H	Branches	Neural rooter	Full neural	Branches + ridge
168	24	522,080	323,576	845,656	522,152
336	48	645,056	397,424	1,042,480	645,200
504	168	915,872	508,616	1,424,488	916,376
512	64	759,808	471,232	1,231,040	760,000

At 512:64, the default ridge rooter has $3H = 192$ coefficients.

Table 10: Adaptation-parameter comparison at the released Cross-RAG 512:64 setting. Frozen backbone parameters are excluded from the first numeric column. External counts follow the released implementations (Ning et al., 2025b; Lee et al., 2026b).

Method	Trainable adaptation params.	$\times$ TS-IFA	Share of backbone
TS-IFA neural	1,231,040	1.00	1.03% of Chronos-2
TS-IFA branches + ridge	760,000	0.62	0.63% of Chronos-2
TS-RAG added modules	4,775,425	3.88	2.33% of Bolt-base
Cross-RAG added modules	8,712,192	7.08	4.25% of Bolt-base
Full horizon ridge baseline, $K = 3$	512	0.00042	0.00025% of Bolt-base
One shared CatBoost gate	$\leq 6,000$ proxy	≤ 0.0049	$\leq 0.005\%$ of Chronos-2

9 Experimental design

9.1 Datasets and model profiles

Table 11: Dataset panels and their roles. All non-date variates are retained for ETT, Weather, and Exchange.

Role	Datasets	Preparation
Primary screen	Electricity, Traffic, Solar, exchange_rate	Hourly Electricity/Traffic; Solar is summed from 10-minute to hourly; Exchange is daily. Electricity drops the nine configured channels.
Full homogeneous panels	ETT_T_1H, ETT_L_1H, ETT_T_15T, ETT_L_15T	Temperature and load are separated. Hourly panels use $P = 24$; 15-minute panels use $P = 96$.
Mixed-quantity ablation	ETTh1, ETTh2, ETTm1, ETTm2, Weather	Original seven-variable ETT panels; Weather is hourly mean aggregation. Cross-variable retrieval is reported only as an ablation.

Table 12: Resolved experiment profiles. A selected winner name fixes family, formula, retrieval space, metric, K , and online mode.

Profile	Datasets	(L, H)	Backbone	Retrieval / methods
test	Electricity	504:168	Chronos-2	raw Euclidean, $K = 3$; smoke-only
screen	four primary	168:24, 336:48, 504:168	Chronos-2	raw/instance Euclidean, $K \in \{1, 3\}$; 10 baselines and 8 gate/reference artifacts
full	primary + four separated ETT	default three	Chronos-2	selected complete screen pipelines only
ultra	same as full	default three	Chronos-2, TabPFN-TS	same winners and output root as full
crossrag	four primary	512:64	Chronos-Bolt	min-max cosine, $K = 15$; one winner vs. released Cross-RAG

Let

$$\mathcal{D}_P = \{\text{Electricity, Traffic, Solar, exchange_rate}\}, \quad \mathcal{S} = \{168:24, 336:48, 504:168\},$$

$$\mathcal{R} = \{\text{raw, instance}\} \times \{\text{Euclidean}\} \times \{K = 1, 3\} \times \{\text{online}\},$$

and let $\mathcal{D}_B$ be the seven stems in Table 5. The 12 = $|\mathcal{D}_P||\mathcal{S}|$ screen settings test, for every $r \in \mathcal{R}$,

$$\mathcal{B}_{\text{screen}} = \{\text{cov_forecast, avgy, y_mean}\} \cup \{\text{<d>_ridge_shared} : d \in \mathcal{D}_B\},$$

$$\mathcal{G}_{\text{screen}} = \bigcup_{A \in \{\text{cov, avgy}\}} \{\text{direct_<A>, bayes_<A>_shared, catboost_<A>_regressor_shared, oracle_<A>_shared}\},$$

where **direct_cov**=**cov_forecast** and **direct_avgy**=**avgy**. Thus the baseline and gate screens contain 10 and 8 artifacts per retrieval pipeline; convex, delta, horizon, classifier, and mixture forms are excluded.

Default publication strides are 24 pooled-adaptation query dates and 128 evaluation query dates; the datastore stride is 24 before period rounding and the cap is 30,000 windows. The test profile uses adaptation/evaluation stride 256, datastore stride 168, and 2,048 store windows. Seed 1 is used throughout.

9.2 Benchmarks and ablations

Table 13: Selected baseline controls and one-family transformations. Retrieval metric is Euclidean and mode is online in every row.

Stem	Space, K	Horizon front	Convex front	Delta front
full	instance, 3	ridge shared / horizon	ridge / convex shared	ridge / delta shared
y	instance, 3	ridge shared / horizon	ridge / convex shared	ridge / delta shared
cov_y	instance, 3	ridge shared / horizon	ridge / convex shared	ridge / delta shared
residual	instance, 3	ridge shared / horizon	ridge / convex shared	ridge / delta shared
cov_avgy	raw, 3	ridge shared / horizon	ridge / convex shared	ridge / delta shared
cov	raw, 3	ridge shared / horizon	ridge / convex shared	ridge / delta shared

Each row retains `<stem>_ridge_shared` as its control. The three fronts replace only the suffix by `ridge_horizon`, `convex_shared`, or `delta_ridge_shared`, respectively.

Table 14: Selected CatBoost controls and one-axis transformations.

A	Space, K	Control	Separate one-axis variants
avgy	instance, 3	regressor shared	classifier shared; regressor shared soft; regressor horizon
cov	instance, 1	regressor shared	classifier shared; regressor shared soft; regressor horizon

For each A , the CatBoost front also includes matching direct, Bayes, and oracle references. No transformation is crossed with another.

Table 15: Root submission fronts and tested model sets.

Front	Model set / controlled axis
screen.slurm	$\mathcal{B}_{\text{screen}}$ and $\mathcal{G}_{\text{screen}}$ on $\mathcal{D}_P \times \mathcal{S} \times \mathcal{R}$
benchmark.slurm	complete selected screen pipelines; full uses Chronos-2, ultra adds TabPFN-TS
k_ablation.slurm	nine selected pipelines, $K \in \{1, 3, 5, 10, 15, 20\}$ only
h_ablation.slurm	selected pipelines, $L = 504$, $H \in \{24, 168, 504\}$ only
l_ablation.slurm	selected pipelines, $H = 24$, $L \in \{24, 168, 504\}$ only
horizon_baselines_ablation.slurm	six controls in Table 13; shared versus horizon ridge
convex_baselines_ablation.slurm	same six; shared ridge versus shared convex
delta_baselines_ablation.slurm	same six; shared ridge versus shared delta-ridge
catboost_ablation.slurm	two controls in Table 14; three separate one-axis variants
mixed_quantity_ablation.slurm	selected pipelines transferred to the five mixed-quantity panels
crossrag.slurm	one selected pipeline versus Cross-RAG at $(L, H, K) = (512, 64, 15)$
ts_ifa\joint\ridge.slurm	joint-gradient ridge TS-IFA; current focus
ts_ifa\joint\neural.slurm	joint-gradient neural TS-IFA
ts_ifa\meta\ridge.slurm	exact support-solve ridge meta-TS-IFA
ts_ifa\meta\neural.slurm	gradient-inner-loop neural meta-TS-IFA
extraction.slurm	standalone extraction smoke or TS-IFA-input front
baselines.slurm	standalone primary-baseline smoke front
gates.slurm	standalone primary-gate smoke front
tables.slurm	standalone result-table front

The completed screen selected six shared-ridge baselines, two hard shared CatBoost regressors, and one shared Bayes gate; no convex or delta baseline was eligible for that screen. Its extraction payloads and controls remain valid. Run the formulation fronts, filter their outputs into an expanded candidate set, then run the K study; `screen.slurm` is not rerun. Publication CatBoost fits allow 300 trees; only the smoke profile uses two.

For setting c , method m , method score $\text{nMSE}_{m,c}$, and matching vanilla $\text{nMSE } v_c$, the reported percentage improvement is

$$I_{m,c} = 100 \frac{v_c - \text{nMSE}_{m,c}}{v_c}.$$

The average table reports the unweighted mean of $I_{m,c}$ over complete settings, not a percentage computed after pooling losses. If n_c is the number of held-out windows in setting c , window-level win rate is

$$W_{m,c} = 100 \frac{1}{n_c} \sum_{i=1}^{n_c} \mathbf{1} \left[\frac{1}{H} \sum_{h=1}^H (G_{i,h} - \hat{G}_{m,i,h})^2 < \frac{1}{H} \sum_{h=1}^H (G_{i,h} - V_{i,h})^2 \right].$$

Final claims use complete held-out T_3 scoring. Optimistic `*_eval_fit` baselines and target-aware oracle gates are appendix diagnostics only.

10 Summary

The project tests a hierarchy of increasing adaptation capacity around frozen forecasting foundation models: direct analogue forecasts, ridge/convex/delta linear families, conditional CatBoost gates, and TS-IFA. Every method shares the same query dates, leakage-free online datastore, query-scale neighbour signals, and held-out T_3 evaluation. This makes capacity, retrieval choice, and validation protocol explicit rather than conflating them with backbone quality.

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